

# Hashing Q&A

## Quadratic Probing and Repeated Probe Indices

### 1 Question 1

#### Question

Given a hash table of size 11 and the hash function

$$h(k) = (3k + 5) \bmod 11,$$

insert the keys

22, 41, 53, 46, 30, 52, 42, 23, 34

using quadratic probing. Show the probe sequence for each key and the final hash table.

#### Quadratic Probing Formula

Use quadratic probing with

$$c_1 = 1, \quad c_2 = 3.$$

Thus

$$h(k, i) = (h(k) + i + 3i^2) \bmod 11, \quad i = 0, 1, 2, \dots$$

#### Probe Sequences

| Key $k$ | $h(k)$ | Probe sequence until insertion |
|---------|--------|--------------------------------|
| 22      | 5      | 5                              |
| 41      | 7      | 7                              |
| 53      | 10     | 10                             |
| 46      | 0      | 0                              |
| 30      | 7      | 7, 0, 10, 4                    |
| 52      | 7      | 7, 0, 10, 4, 4, 10, 0, 7, 9    |
| 42      | 10     | 10, 3                          |
| 23      | 8      | 8                              |
| 34      | 8      | 8, 1                           |

#### Final Hash Table

| Index | 0  | 1  | 2 | 3  | 4  | 5  | 6 | 7  | 8  | 9  | 10 |
|-------|----|----|---|----|----|----|---|----|----|----|----|
| Key   | 46 | 34 | – | 42 | 30 | 22 | – | 41 | 23 | 52 | 53 |

#### Write This in Exam

Use

$$h(k, i) = (h(k) + i + 3i^2) \bmod 11.$$

Then the final table is

| $i$    | 0  | 1  | 2 | 3  | 4  | 5  | 6 | 7  | 8  | 9  | 10 |
|--------|----|----|---|----|----|----|---|----|----|----|----|
| $T[i]$ | 46 | 34 | — | 42 | 30 | 22 | — | 41 | 23 | 52 | 53 |

with probe sequences listed in the table above.

## 2 Question 2

### Question

Given a hash table of size

$$m = 4013$$

and the hash function

$$h(k) = (3k + 5) \bmod 4013$$

with quadratic probing, find a pair of consecutive probe indices  $i$  and  $j$  with

$$0 \leq i, j \leq 4013, \quad i \neq j,$$

that produce the same table index.

### Answer

Take

$$i = 1337, \quad j = 1338.$$

These are consecutive probe indices.

### Proof

For quadratic probing,

$$h(k, i) = (h(k) + i + 3i^2) \bmod 4013.$$

For consecutive probe indices  $i$  and  $i + 1$ , the difference between the quadratic offsets is

$$(i + 1) + 3(i + 1)^2 - (i + 3i^2) = 6i + 4.$$

We need

$$6i + 4 \equiv 0 \pmod{4013}.$$

Taking

$$i = 1337$$

works because

$$6(1337) + 4 = 8026 = 2 \cdot 4013.$$

Thus

$$1337 + 3(1337)^2 \equiv 1338 + 3(1338)^2 \pmod{4013}.$$

Therefore

$$h(k, 1337) = h(k, 1338)$$

for every key  $k$ . The consecutive probe indices 1337 and 1338 produce the same table index.

**Write This in Exam**

With  $c_1 = 1$  and  $c_2 = 3$ ,

$$h(k, i) = (h(k) + i + 3i^2) \bmod 4013.$$

For consecutive probes  $i$  and  $i + 1$ , the offset difference is

$$6i + 4.$$

Choose  $i = 1337$  and  $j = 1338$ . Since

$$6(1337) + 4 = 8026 = 2 \cdot 4013,$$

the two offsets are congruent modulo 4013. Hence the two consecutive probe indices produce the same table index.